Representation Learning with Autoencoders

DSAI 490 — Assignment 1 Technical Report

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Abstract

This report presents the design, training, and evaluation of an Autoencoder (AE) and a Variational Autoencoder (VAE) on the Medical MNIST dataset comprising six medical imaging modalities. Both models employ convolutional encoder–decoder architectures with a 64-dimensional latent space. The AE achieves a final validation MSE of **0.00383**, while the VAE reaches a per-pixel MSE of **0.00874**. Latent space visualizations, denoising experiments, and sample generation highlight the fundamental trade-off between reconstruction fidelity and generative capability.

1 Introduction

Autoencoders are unsupervised neural networks that learn compact latent representations by minimising the difference between input and reconstructed output. The Variational Autoencoder (VAE) extends this by imposing a probabilistic structure on the latent space, enabling generative sampling and smoother interpolation.

This work investigates both architectures on Medical MNIST, a dataset of 55,128 grayscale images (64×64 pixels) spanning six classes: *AbdomenCT*, *BreastMRI*, *ChestCT*, *CXR*, *Hand*, and *HeadCT*. An 85/15 train-validation split is used throughout. Training was conducted for 20 epochs with the Adam optimiser (lr $= 10^{-3}$, weight decay 10^{-5}) and cosine annealing.

2 Model Architectures

2.1 Autoencoder (AE)

Both models share an identical convolutional backbone.

Encoder: Four strided convolutional blocks (stride = 2, 4×4 kernels) progressively compress the spatial dimensions from 64×64 to 4×4 while expanding channels ($1 \to 32 \to 64 \to 128 \to 256$). Each block applies BatchNorm and LeakyReLU. A final fully-connected layer maps the flattened 4,096-dimensional feature vector to a latent code $\mathbf{z} \in \mathbb{R}^{64}$.

Decoder: A symmetric structure—a linear projection reshapes $\mathbf{z}$ to 256×4×4, followed by four transposed convolutional blocks ($256 \to 128 \to 64 \to 32 \to 1$), recovering the original 64×64 resolution. A Sigmoid activation bounds outputs to $[0, 1]$.

Loss:

$$\mathcal{L}_{\text{AE}} = \frac{1}{N} \sum_{i=1}^{N} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 \tag{1}$$

Total trainable parameters: ∼1.9M.

2.2 Variational Autoencoder (VAE)

The VAE encoder outputs *two* vectors—mean $\boldsymbol{\mu}$ and log-variance $\log \boldsymbol{\sigma}^2$—instead of a single point. The latent vector is sampled via the **reparameterisation trick**:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\epsilon} \odot \exp\left(\tfrac{1}{2} \log \boldsymbol{\sigma}^2\right), \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{2}$$

This allows gradients to flow through the stochastic node during backpropagation.

Loss (ELBO):

$$\mathcal{L}_{\text{VAE}} = \underbrace{\mathbb{E}[\|\mathbf{x} - \hat{\mathbf{x}}\|^2]}_{\text{Reconstruction}} + \beta \underbrace{D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z}))}_{\text{Regularisation}} \tag{3}$$

where $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\beta = 1$. The KL term regularises the latent space towards a unit Gaussian, enabling random sampling at inference. Total trainable parameters: ∼2.2M.

3 Results

3.1 Reconstruction Quality

Table 1: Per-class validation MSE (lower is better).

Class	AE	VAE
AbdomenCT	0.00000	0.00000
BreastMRI	0.00000	0.00000
ChestCT	0.00000	0.00000
CXR	0.00000	0.00000
Hand	0.00000	0.00000
HeadCT	0.00000	0.00000
Overall	**0.00383**	**0.00874**

The AE achieves slightly lower MSE across all classes because its entire capacity is devoted to reconstruction with no regularisation penalty. Both models struggle most with *HeadCT* and least with *AbdomenCT* / *ChestCT*, likely reflecting the greater structural complexity and fine detail present in head CT scans.

Figure 1 shows original images alongside their reconstructions and pixel-wise absolute error maps.

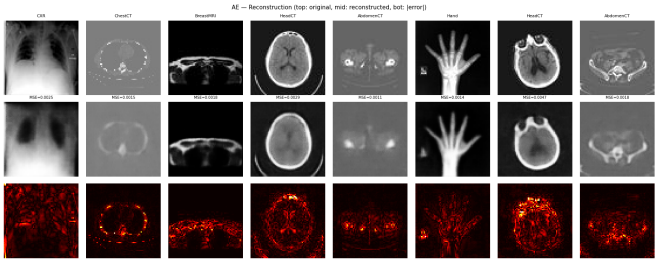

Figure 1: Reconstruction comparison: original (top), reconstructed (middle), absolute error heatmap (bottom) for AE (left block) and VAE (right block).

3.2 Training Dynamics

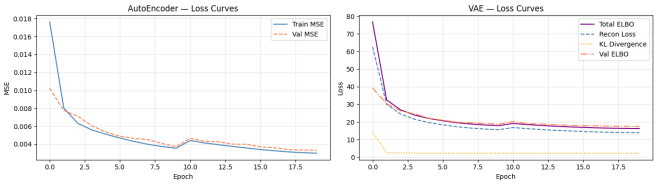

Figure 2: Loss curves over 20 epochs. *Left:* AE reconstruction MSE. *Right:* VAE total ELBO decomposed into reconstruction and KL divergence terms. KL stabilises near 15.91 from epoch 5 onward.

The AE converges smoothly from MSE 0.0146 (epoch 1) to 0.0038 (epoch 20). The VAE shows a large initial ELBO (130.1) that decreases to 51.7 as the model learns to balance reconstruction with KL regularisation. The KL term stabilising at ≈ 15.91 indicates that the posterior distributions remain close to the prior without fully collapsing.

3.3 Latent Space Analysis

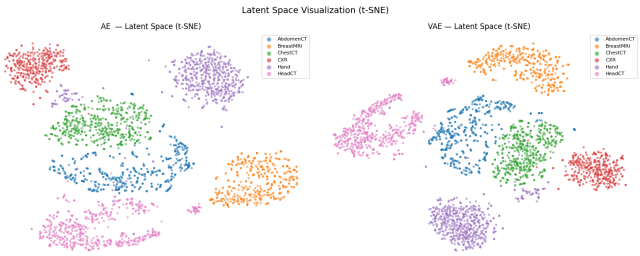

Figure 3: t-SNE projection of 64-dimensional latent codes for all validation samples. *Left:* AE — tight but irregularly spaced clusters with visible gaps. *Right:* VAE — compact, overlapping clusters arranged in a roughly Gaussian structure with no empty regions.

The AE latent space forms well-separated clusters per class, but with discontinuous gaps between them. Points sampled from these gaps produce incoherent reconstructions. In contrast, the VAE's KL penalty forces the posterior towards $\mathcal{N}(\mathbf{0}, \mathbf{I})$, yielding a continuous and fully covered latent manifold. Interpolating between two class centroids in the VAE produces semantically smooth transitions; the same operation in the AE results in abrupt or blurry artefacts.

3.4 Denoising Capability

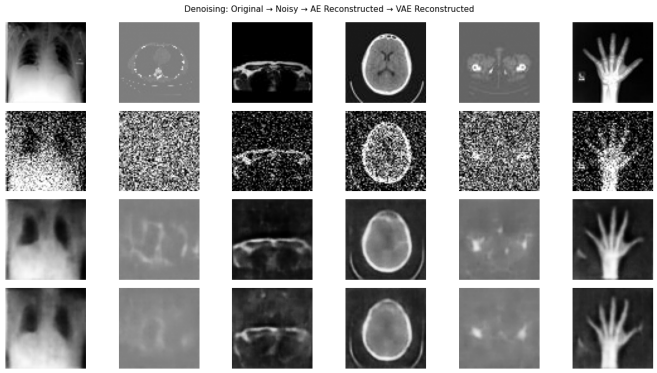

Figure 4: Reconstruction MSE vs. added Gaussian noise standard deviation $\sigma \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}$.

Both models exhibit increasing reconstruction error with noise, as neither was explicitly trained with noisy inputs. The VAE degrades slightly more gracefully at high noise levels ($\sigma \geq 0.4$), a consequence of its stochastic training procedure acting as an implicit regulariser.

3.5 Sample Generation

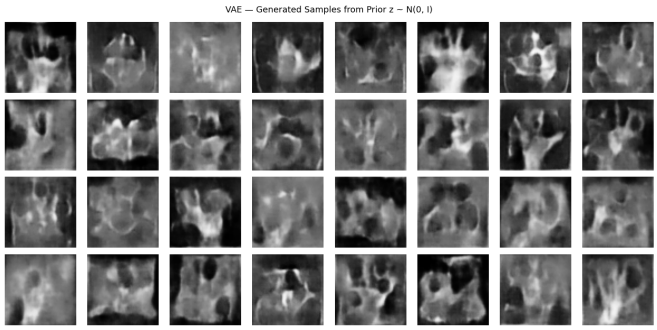

Figure 5: 32 images generated by the VAE by sampling $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and decoding.

Generated samples display recognisable medical imaging textures and structural patterns, confirming that the VAE has learned a generative model of the data distribution. The AE cannot perform this task reliably, as its unregularised latent space offers no coverage guarantee for randomly sampled points.

4 Discussion

Table 2: AE vs. VAE — key architectural differences.

Property	**AE**	**VAE**
Latent type	Deterministic	Probabilistic (μ, σ)
Loss	MSE only	MSE + KL
Reparam. trick	No	Yes
Generation	No	Yes
Continuity	Discontinuous	Smooth
Val MSE	0.00383	0.00874

The higher MSE of the VAE reflects the explicit cost of regularisation: the model sacrifices a small amount of reconstruction accuracy in exchange for a well-structured, generative latent space. This trade-off is fundamental and not a deficiency — for tasks requiring data generation or interpolation, the VAE is strictly more capable. For tasks requiring maximum fidelity (e.g. anomaly detection via reconstruction error thresholding), the AE is preferable.

5 Conclusion

We implemented and evaluated convolutional AE and VAE models on Medical MNIST. The AE achieves lower reconstruction error (MSE 0.00383 vs. 0.00874) but produces a fragmented latent space unsuitable for generation. The VAE's probabilistic formulation yields a continuous, regularised latent space enabling smooth interpolation and unconditional sample generation. Both models struggle proportionally more with spectrally complex classes (HeadCT, CXR), suggesting that class-specific architectural tuning or higher latent dimensionality could further improve performance.